ENGINEERING INNOVATION PROJECT REPORT

on

SentinEL: Adversarial Robustness Analysis of ML-Based Phishing URL Detection

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INDRASHIL UNIVERSITY

School of Engineering — Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the **Engineering Innovation Project** work entitled **"SentinEL: Adversarial Robustness Analysis of ML-Based Phishing URL Detection"** by Mourya Reddy Udumula (Enrollment No. 220110101055) of Indrashil University, submitted towards the partial fulfillment of requirements of the degree of Bachelor of Technology — Computer Science and Engineering, during the academic year 2025–26. The results and findings contained in this project have not been submitted in part or full to any other University or Institute for the award of any other Degree or Diploma.

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ABSTRACT

Phishing attacks remain the most prevalent entry vector for targeted cyberattacks globally. Static URL blacklists and rule-based heuristics fail to detect novel, zero-day phishing infrastructure, and the published machine learning detection literature evaluates classifiers exclusively on clean, unperturbed inputs — creating an empirically unjustified confidence in ML-based phishing defences. This project presents SentinEL (Security ENTinel through Interpretable and Explainable Learning), a research-grade adversarial robustness analysis system that quantifies how ML-based phishing URL detection degrades under realistic Unicode-based evasion.

The system is built around a Random Forest classifier (n_estimators=100, max_depth=None) trained on 11,000 URLs sourced from PhishTank and a verified legitimate URL corpus, using a 17-dimensional feature vector derived from URL structure and lexical properties. The dataset was curated and quality-validated by Mourya Reddy Udumula, including cross-validation of a 500-URL sample against VirusTotal for label accuracy. Baseline classification accuracy is 97.2% on a 2,200-URL stratified hold-out test set.

The core research contribution is a systematic adversarial evaluation: homoglyph substitution attacks (19 Latin-to-Cyrillic Unicode character pairs applied exclusively to URL domain and path components via urlparse/urlunparse) reduce classifier accuracy from 97.2% to 81.4% — a 15.8 percentage point degradation. This quantifies a critical vulnerability absent from all published URL-based phishing detection benchmarks. Feature sensitivity analysis reveals that string-level lexical features (Shannon entropy, keyword frequency) are the primary adversarial attack surface, while metadata features (domain age, SSL validity) show zero sensitivity.

A three-method explainability benchmark establishes: Gini Impurity (< 0.001ms), SHAP TreeExplainer (~0.23ms), and LIME TabularExplainer (130ms). Gini and SHAP meet the 2ms SOC real-time threshold; LIME does not. The operational pipeline (Jeet Upadhyaya Anand) achieves 5× throughput improvement via ThreadPoolExecutor-based concurrent IOC feed ingestion. All findings were presented at the Indrashil University Annual Research Symposium 2025.

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Chapter 1: Introduction

1.1 Project Summary

SentinEL (Security ENTinel through Interpretable and Explainable Learning) is a research-grade adversarial robustness analysis system for machine learning-based phishing URL detection. The project investigates a critical and largely unexplored security question: when ML-based phishing detectors achieve high accuracy on clean inputs, how resilient are they against adversarially crafted inputs designed to evade classification?

The core system is a Random Forest classifier (100 estimators, unlimited depth) trained on 11,000 URLs sourced from PhishTank and a verified legitimate URL corpus, using a 17-dimensional feature vector derived exclusively from URL structure and lexical properties — requiring no live network calls at classification time. The dataset was curated and quality-validated as part of this project, including cross-validation of a 500-URL sample against VirusTotal for label accuracy. The baseline classifier achieves 97.2% accuracy on a 2,200-URL stratified hold-out test set, matching published state-of-the-art results for URL-based phishing detection.

The primary research contribution extends beyond baseline detection. SentinEL systematically subjects the classifier to homoglyph substitution attacks — replacing 19 ASCII Latin characters with visually identical Cyrillic Unicode equivalents applied exclusively to URL domain and path components via urlparse/urlunparse — and quantifies the resulting degradation: accuracy drops from 97.2% to 81.4%, a 15.8 percentage point reduction. This finding demonstrates that even high-performing phishing classifiers harbour exploitable Unicode-based vulnerabilities that standard clean-input benchmarking will never reveal.

A secondary research contribution is a comparative benchmark of three explainability methods — Gini Impurity attribution (< 0.001ms), SHAP TreeExplainer (~0.23ms), and LIME TabularExplainer (130ms) — establishing empirical latency boundaries for real-time SOC deployment. The project is implemented as a fully modular Python codebase with 10 pytest unit tests (all passing in 1.82s), SHA-256 model integrity verification, and GitHub Actions CI. Findings were presented at the Indrashil University Annual Research Symposium 2025.

1.2 Project Scope

SentinEL's scope is focused on three research concerns: adversarial vulnerability quantification, explainability benchmarking, and operational pipeline design. Two datasets coexist in the repository: the full 11,000-URL research corpus (data/phishing_dataset.csv), which underlies all quantitative results reported in this document, and a 14-row demonstration sample (data/demo_sample.csv) included for CI/CD test execution without requiring the full dataset download. All experimental findings in Chapters 5 and 6 are derived exclusively from the full research corpus.

In-Scope:

- Dataset curation: 11,000 URLs from PhishTank and verified legitimate corpus, including VirusTotal label validation and EDA; 80/20 stratified train-test split
- Feature engineering: 17 URL-derived features computable without network calls — Shannon entropy, URL length, TLD risk score, subdomain count, domain age, SSL state, @ symbol, IP-as-host, URL shortener, keyword frequency, dot count, hyphen-in-domain, special char count, path depth, query parameter count, HTTPS flag, domain token count
- Model training: Random Forest with 5-fold cross-validated hyperparameter configuration
- Adversarial evaluation: 19-pair Latin-to-Cyrillic homoglyph substitution on domain and path only; feature sensitivity analysis identifying which features degrade under attack
- Explainability benchmarking: latency comparison of Gini, SHAP, and LIME against 2ms SOC real-time threshold across 100 samples
- Operational pipeline: ThreadPoolExecutor-based concurrent IOC feed ingestion achieving 5× throughput improvement (Jeet Upadhyaya Anand)
- Test coverage: 10 pytest unit tests across all core modules, GitHub Actions CI

Out-of-Scope:

- Browser extension or SIEM production integration
- Continuous model retraining from live threat feeds
- Visual similarity (screenshot/pHash) or DOM content-based detection

- Attack simulation against real infrastructure
- Large-scale cloud or enterprise deployment

1.3 Project Objectives

- Design and implement a Random Forest phishing URL classifier achieving >97% baseline accuracy using a 17-feature engineering approach that requires no live network calls at inference time
- Conduct a systematic adversarial robustness evaluation under homoglyph substitution attack, quantifying accuracy degradation and identifying adversarially sensitive features within the 17-dimensional feature vector
- Benchmark Gini Impurity, SHAP TreeExplainer, and LIME TabularExplainer against a 2ms SOC real-time latency threshold, providing empirical guidance for XAI method selection in security operations
- Design a fully modular Python architecture cleanly separating ML research (feature extraction, classification, explainability, adversarial evaluation) from operational pipeline concerns (IOC ingestion, alert dispatching, audit logging)
- Achieve comprehensive pytest-based test coverage across all core modules with SHA-256 model integrity verification and GitHub Actions CI
- Document all findings reproducibly with JSON result artifacts and a technical report, supporting the principle that adversarial robustness must be measured — not assumed — in security-critical ML systems

1.4 Background and Motivation

Phishing attacks represent the most prevalent and economically damaging form of social engineering cybercrime. The Anti-Phishing Working Group (APWG) consistently reports millions of unique phishing URLs detected annually, with attack volume continuing to grow year-on-year. Phishing serves as the entry vector for over 90% of targeted cyberattacks, making URL-level detection a foundational security problem.

Traditional defences — primarily static URL blacklists maintained by services such as Google Safe Browsing — are fundamentally reactive. The average lifespan of a phishing site is under 24 hours, meaning blacklists detect and block URLs only after the initial wave of victims has been compromised. This temporal gap is the critical structural vulnerability of conventional defence architectures. Machine learning-based detection offers a proactive alternative: classifiers trained on the statistical patterns of phishing infrastructure generalise to novel, zero-day URLs that have never appeared on any blacklist.

However, a fundamental and systematically underexamined vulnerability exists in all published ML-based phishing detection work: adversarial robustness. Published accuracy figures — including state-of-the-art results exceeding 97–99% — are measured exclusively on clean, unperturbed inputs. No published phishing detection paper, to the authors' knowledge, evaluates performance under systematic Unicode-based evasion.

Homoglyph attacks exploit Unicode's inclusion of visually identical characters from multiple scripts. The Latin 'a' (U+0061) and Cyrillic 'а' (U+0430) are pixel-identical at standard browser rendering resolutions, yet have completely different byte representations that fundamentally alter string-level features. An attacker who understands that a classifier relies on keyword frequency and Shannon entropy can craft URLs that evade these features while remaining fully functional for credential harvesting. The motivation for SentinEL is to close this measurement gap: not merely to build a phishing detector, but to systematically quantify its breaking point, identify adversarially sensitive features, and benchmark the explainability tools that allow security analysts to audit decisions — prerequisites for trustworthy deployment under the EU AI Act's transparency requirements that enter enforcement in August 2026.

1.5 Literature Review

Phishing detection is a mature research domain with over two decades of academic and industry development. The following review analyses existing approaches across five criteria directly relevant to SentinEL's research focus: detection accuracy on clean inputs, inference latency, adversarial robustness testing, explainability provision, and deployability in real-time security operations. This multi-criteria evaluation reveals the specific research gap that SentinEL addresses.

1.5.1 Static and List-Based Methods

URL blacklisting (Google Safe Browsing, PhishTank, OpenPhish) stores hashes of known malicious URLs for fast lookup. While achieving near-zero false positive rates on known phishing sites, these systems fail entirely against zero-day URLs. Marchal et al. (2014) quantify this temporal gap in their analysis of PhishStorm data: the average phishing site lifetime is under 24 hours, meaning blacklists provide protection only after the initial victim wave. Adversarial robustness is structurally irrelevant for blacklist-based systems — any URL not previously seen evades them by definition, regardless of how similar it is to known phishing patterns. No adversarial evaluation has been published for these systems because the concept does not meaningfully apply.

1.5.2 Heuristic and Rule-Based Methods

CANTINA (Zhang et al., 2007) pioneered content-based heuristics using TF-IDF analysis of page text to detect phishing. Rule-based browser plugins (SpoofGuard, PwdHash) apply fixed pattern matching against suspicious URL characteristics. These methods achieve 90–95% accuracy in their original evaluations but degrade as attackers learn and adapt to the specific rules. More critically, their adversarial robustness is effectively zero by design — rules are static, finite, and publicly reverse-engineerable. A determined attacker needs only enumerate the rule set once to craft a compliant evasion strategy permanently.

1.5.3 Feature-Engineered Machine Learning

Jain and Gupta (2019) achieve 97.7% accuracy on a UCI Phishing Dataset using Random Forest with 30 URL and HTML features — the closest published baseline to SentinEL's approach. Mohammad et al. (2015) report 97.1% with a similar feature set. These are genuine achievements in clean-input detection. However — and this is the critical observation — none of these published evaluations includes any form of adversarial robustness testing. Accuracy figures are measured exclusively on clean, unperturbed test sets, producing what Apruzzese et al. (2022) identify as an systematically optimistic view of real-world security ML performance.

The choice of Random Forest for SentinEL follows directly from this literature. Random Forest handles heterogeneous feature types without normalisation, provides native Gini Impurity

attribution compatible with real-time SOC latency requirements, generates hundreds of independent decision boundaries that an adversary must simultaneously satisfy to achieve successful evasion, and operates within the sub-100ms inference latency constraint of operational deployment.

1.5.4 Deep Learning Approaches

URLTran (Maneriker et al., 2021) applies transformer-based character-level URL encoding, achieving competitive accuracy without manual feature engineering. CNN-based approaches (Le et al., 2018) report 98.3% on benchmark URL datasets. However, Apruzzese et al. (2022) demonstrate a counterintuitive finding: deep learning cybersecurity classifiers are generally more vulnerable to adversarial perturbation than classical ML ensembles, because their high-dimensional learned representations contain more exploitable directions in feature space. The inference latency of transformer models (100–500ms) also disqualifies them from real-time SOC deployment at scale.

1.5.5 Comparative Analysis and Research Gap Identification

The following table compares phishing detection approaches across the five evaluation criteria defined above. This comparison reveals SentinEL's specific contribution to the research landscape.

Approach	Best ReportedAccuracy	InferenceLatency	Adversarial Robustness Tested	Explainability	SOCDeployable
Blacklisting	~100% known	<1ms	N/A	None	Partial
Heuristic/Rules	90–95%	<10ms	No	Partial	Yes
Feature-Eng. ML (RF)	97–99%	10–100 ms	No	Feature importance	Yes

Deep Learning (CNN/Transformer)	97–99%	100–500ms	Rarely	Black-box	Limited
SentinEL (this work)	97.2% clean81.4% attacked	~12ms	YES — 15.8pp quantified	Gini+SHAP+LIME benchmarked	Yes

Table 1.1: Comparison of Phishing Detection Approaches. The table reveals that SentinEL is the only approach that explicitly measures and reports adversarial robustness alongside baseline accuracy. The literature systematically reports only clean-input performance — SentinEL's primary contribution is making the adversarial vulnerability visible and quantified, establishing the first published benchmark for homoglyph substitution attack against URL-based Random Forest classifiers.

1.5.6 Explainability Gap in Security Operations

Ribeiro et al. (2016) introduced LIME and Lundberg and Lee (2017) introduced SHAP as general-purpose post-hoc explanation tools. Both have been applied to security classifiers in the academic literature. However, no published work benchmarks their operational latency in a phishing detection context against a real-time SOC deployment threshold. Security operations centres process thousands of URL classifications per hour; an explanation method that requires 130ms per sample introduces unacceptable throughput degradation at that scale. SentinEL's secondary contribution — establishing empirically that Gini and SHAP meet the ≤ 2ms threshold while LIME does not — fills this operational gap in the literature and provides directly actionable guidance for XAI tool selection in SOC environments.

1.6 Methodology Overview

Phase	Description	Key Tools	Output Artifact

1. Data Engineering	Acquire 11,000 URLs (PhishTank + legitimate corpus). VirusTotal label validation. 17-feature engineering. EDA. 80/20 stratified split.	pandas, numpy, PhishTank, Cisco Umbrella	data/phishing_dataset.csv
2. Model Training	Train Random Forest. 5-fold cross-validated configuration. Evaluate on hold-out test set. Compute SHA-256 integrity checksum.	scikit-learn	sentinel_ultima.pkl + model_checksum.sha256
3. Adversarial Evaluation	Apply 19-pair homoglyph substitution to domain/path only. Re-extract features. Measure accuracy degradation. Feature sensitivity diagnostic.	attack_generator.py, homoglyph_map.py	results_adversarial.json
4. Explainability Benchmark	Measure median attribution latency across 100 samples for Gini, SHAP, LIME. Compare against 2ms SOC threshold.	shap, lime, GiniExplainer	results_explainability.json
5. Operational Pipeline	Concurrent IOC feed ingestion via ThreadPoolExecutor. Real-time alert dispatch. Forensic audit logging to SQLite.	ThreadPoolExecutor, SQLite, pathlib	audit.db + alert stream

1.7 Summary

This chapter introduced SentinEL: its research motivation, scope, objectives, literature positioning, and methodology. The project occupies a specific and currently unoccupied niche in

the phishing detection research landscape — moving beyond accuracy benchmarking to adversarial vulnerability quantification with explainable, auditable attributions. The comparative analysis in Table 1.1 demonstrates that no existing published system simultaneously achieves (a) systematic adversarial evaluation, (b) real-time XAI benchmarking, and (c) modular architecture enabling clear research attribution. Subsequent chapters detail system requirements (Chapter 2), system analysis and module design (Chapter 3), architectural design (Chapter 4), complete implementation with code (Chapter 5), and experimental conclusions with future work (Chapter 6).

Chapter 2: System Requirement Study

2.1 Hardware and Software Characteristics

Efficient operation of SentinEL depends on hardware and software configurations that support ML training, adversarial evaluation, and explainability benchmarking. SHAP TreeExplainer and LIME TabularExplainer benefit from multi-core parallelism during benchmark runs; the ThreadPoolExecutor-based operational pipeline similarly benefits from multi-core availability.

2.1.1 Hardware Requirements

Specification	Minimum (Inference Only)	Recommended (Full Training + Evaluation)
Processor	Intel Core i3 (8th Gen) / AMD Ryzen 3	Intel Core i5 (10th Gen)+ / AMD Ryzen 5+, multi-core
RAM	4 GB (feature extraction and inference only)	8 GB+ (SHAP/LIME concurrent benchmark runs)
Storage	500 MB (model + dataset + virtualenv)	2 GB+ (full virtualenv, dataset, JSON outputs)
OS	Windows 10/11, Ubuntu 20.04+, macOS Catalina+	Ubuntu 22.04 LTS recommended for CI/CD consistency
Network	None (inference requires no live calls)	Required for PhishTank dataset download only

2.1.2 Software Requirements

Component	Version	Purpose
Python	3.8+	Core runtime
scikit-learn	1.0+	RandomForestClassifier, cross-validation, metrics

pandas	1.3+	Dataset loading, EDA, feature DataFrame operations
numpy	1.21+	17-dimensional feature vector operations
shap	0.41+	SHAP TreeExplainer — explainability benchmarking
lime	0.2.0+	LIME TabularExplainer — explainability benchmarking
pycryptodome	3.15+	SHA-256 model integrity verification
pytest	7.0+	Unit test framework — 10 tests, 1.82s runtime
fpdf2	2.7+	docs/SentinEL_Technical_Report.pdf generation
concurrent.futures	stdlib	ThreadPoolExecutor for operational pipeline

2.2 Functional Requirements

ID	Requirement	Rationale	Implementation
FR-01	Extract exactly 17 URL-derived security features from any valid URL string	Provides the complete feature vector for classifier input; 17 features were selected to balance accuracy with computational tractability	feature_extractor.py: FeatureExtractor.extract()
FR-02	Classify a 17-feature vector as phishing (1) or legitimate (0) with confidence probability	Binary classification with confidence enables threshold calibration for different deployment risk tolerances	classifier.py: PhishingClassifier.predict() + predict_proba()
FR-03	Reject None, non-string, and empty inputs with TypeError	Security systems must not silently fail on malformed input; type guards prevent	Input validation in all public module interfaces

	or ValueError before any ML computation	downstream errors and potential exploitation	
FR-04	Apply homoglyph substitution exclusively to domain and path components — scheme always preserved	Naive full-string substitution produces syntactically invalid URLs (Cyrillic in 'http://') trivially detectable by parsers, invalidating the adversarial experiment	attack_generator.py: apply_homoglyphs_to_url() via urlparse/urlunparse
FR-05	Measure and report attribution latency for Gini, SHAP, and LIME across 100 samples	Establishes empirical evidence for XAI method selection against the 2ms SOC real-time threshold	experiments/explainability_bench.py → results_explainability.json
FR-06	Verify model file integrity via SHA-256 checksum before every classification load	Detects model file corruption between training and deployment; supports audit trail for model provenance	baseline_eval.py: verify_model_integrity()
FR-07	Ingest high-velocity IOC feeds with concurrent threading	Sequential IOC ingestion creates throughput bottlenecks at operational IOC volumes (thousands per hour)	pipeline/pipeline.py: ThreadPoolExecutor(max_workers=N)

FR-08	Serialise all experiment results to JSON with research_context fields	Enables reproducibility, cross-experiment comparison, and self-documenting scope limitations	results_baseline.json, results_adversarial.json, results_explainability.json

2.3 Non-Functional Requirements

ID	Requirement	Target	Measured Result	Verified By
NFR-01	Single-URL inference latency	< 100ms	~12ms	experiments/baseline_eval.py timing
NFR-02	Gini attribution latency (SOC real-time)	< 2ms	< 0.001ms	results_explainability.json
NFR-03	SHAP attribution latency (SOC real-time)	< 2ms	~0.23ms	results_explainability.json
NFR-04	Baseline classification accuracy	> 97%	97.2%	results_baseline.json
NFR-05	Pytest suite execution time	< 30s	1.82s (10 tests)	pytest --tb=short output
NFR-06	Pipeline throughput vs. sequential	≥ 3× improvement	5× latency reduction	pipeline/pipeline.py benchmark

2.4 Security and Integrity Requirements

Requirement	Technical Challenge	SentinEL Implementation	Documented Limitation
Model Integrity Verification	Python pickle files execute code on load — a tampered or corrupted model could silently poison classification results	SHA-256 checksum computed at training time and stored as model_checksum.sha256; verified before every load	Does not protect against an adversary with simultaneous write access to both the model file and the checksum file — documented explicitly in baseline_eval.py docstring
Input Sanitisation	None, type-incorrect, or structurally malformed inputs propagate into feature extraction causing runtime errors or incorrect outputs	Explicit TypeError/ValueError guards as the first operation in all public module functions — before any URL parsing or ML computation	Validates type and non-null; does not currently validate URL syntactic structure (schema, TLD format)
URL-Safe Adversarial Scope	Applying homoglyphs to http:// produces syntactically invalid URLs — parsers reject them before classification, making the	urlparse() decomposes URL into components; substitution applied only to netloc and path; urlunparse() reconstructs valid URL with preserved scheme	Attack limited to domain/path; query strings and fragments not attacked (potential extension for future work)

	attack trivially non-threatening		
Model File Privacy	Binary .pkl files committed to version control permanently increase repository size and expose model internals	data/*.pkl listed in .gitignore; data/README.md documents runtime regeneration via python experiments/baseline_eval.py	Regeneration requires the full dataset — documented as a prerequisite in README.md

2.5 Use Case Analysis

Primary Actor — Security Researcher: Executes three experiment scripts sequentially. (1) python experiments/baseline_eval.py: trains or loads model, verifies SHA-256 integrity, evaluates 97.2% baseline accuracy, writes results_baseline.json. (2) python experiments/adversarial_eval.py: applies 19-pair homoglyph attack, re-evaluates accuracy (81.4%), runs feature sensitivity diagnostic, writes results_adversarial.json with research_context field. (3) python experiments/explainability_bench.py: benchmarks Gini/SHAP/LIME latency across 100 samples, writes results_explainability.json.

Secondary Actor — SOC Analyst: Submits a URL to pipeline/pipeline.py for concurrent feature extraction and classification. Receives binary prediction (phishing/legitimate), confidence score, and GiniExplainer top-5 feature attribution — all within 2ms attribution latency. Every classification is logged to the audit ledger (ops/audit_ledger.py) for forensic traceability.

Tertiary Actor — System Administrator: Manages model versioning via the MODEL_VERSION table in the audit ledger, reviews classification logs in the QUERY table for post-incident forensic analysis, and triggers model file regeneration when updates are required.

2.6 Process Flow Analysis

1. URL string provided as input — type validated (TypeError if None or non-string)

2. FeatureExtractor.extract() produces 17-element NumPy array — all numeric, no missing values
3. PhishingClassifier.predict() produces binary label; predict_proba() produces confidence [0,1]
4. apply_homoglyphs_to_url() applies Cyrillic substitution to netloc and path only
5. Feature extraction repeated on attacked URL — produces attacked feature vector
6. PhishingClassifier.predict() on attacked features — accuracy degradation measured as delta
7. GiniExplainer.explain_gini() returns top-5 feature attribution in < 0.001ms
8. All results serialised to JSON with timestamps, methodology notes, and research_context fields

2.7 Data Flow Analysis

Input Flow: Raw URL string → FeatureExtractor.extract() → (17,) NumPy array → PhishingClassifier → Binary label + confidence

Adversarial Flow: Raw URL → urlparse() decomposition → Cyrillic substitution on netloc+path → urlunparse() reconstruction → FeatureExtractor.extract() → Attacked feature vector → PhishingClassifier → Attacked accuracy delta

Explainability Flow: (17,) feature array + feature name list → GiniExplainer/SHAPExplainer/LIMEExplainer → Attribution dictionary {feature_name: importance_score} + latency measurement → results_explainability.json

Operational Flow: IOC feed → ThreadPoolExecutor.map(extract_and_classify) → Concurrent feature extraction across N workers → Classification results → Alert dispatcher (async) → audit_ledger.log_to_audit() → SQLite QUERY table

2.8 Technical Feasibility Analysis

The project is fully technically feasible using a mature, stable, and well-documented open-source stack. Python's scikit-learn is the industry standard for classical ML and has been in production

use since 2007. SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) are the two most widely peer-reviewed XAI libraries in security research. All components are open-source, version-pinnable via requirements.txt, and executable on commodity hardware without GPU requirements. The most complex implementation decisions — urlparse-based URL-safe attack scoping, character-level entropy measurement to prevent encoding artefacts, and the GiniExplainer unified interface — were validated within the project timeline as evidenced by the passing pytest suite.

2.9 User Requirement Analysis

The primary user requirement is reproducibility and transparent attribution. Security researchers require (a) a modular codebase where each concern lives in its own module with a well-defined public API, (b) comprehensive documentation (README, docstrings, technical report), and (c) verifiable results (pytest tests, JSON artifacts, SHA-256 model checksum). CONTRIBUTORS.md explicitly attributes which modules belong to Mourya Reddy Udumula (dataset curation, ML architecture, adversarial research, explainability benchmarking) and Jeet Upadhyaya Anand (operational pipeline, audit logging), enabling unambiguous academic attribution — a prerequisite for co-authored undergraduate research submissions.

Secondary users (SOC analysts) require latency measurements directly comparable against their operational SLA. The benchmark output in results_explainability.json provides method names, median latencies, ratios relative to Gini baseline, and an explicit 2ms threshold field — designed specifically for comparative tool-selection use cases without requiring the analyst to interpret raw numbers.

2.10 System Architecture Requirements

- Core research modules (feature_extractor.py, classifier.py, explainer.py) must be independently importable and testable without pipeline or adversarial module dependencies
- Adversarial modules (adversarial/homoglyph_map.py, adversarial/attack_generator.py) isolated in a subpackage separating attack logic from classification and feature logic

- Experiment scripts (experiments/*.py) are top-level entry points importing from core modules; all guarded with if __name__ == '__main__' to prevent accidental execution on import
- Operational pipeline (pipeline/pipeline.py) isolated from research modules — imports classifier.py for inference but does not depend on adversarial/ or explainability/ components, enabling independent deployment
- All test classes in tests/test_sentinEL.py map 1:1 to core module concerns for clear test coverage attribution

2.11 Testing Requirements

Test Class	Tests	Coverage Focus	Boundary Conditions
TestFeatureExtractor	3	Output shape, None rejection, numeric-only output	len(features)==17; TypeError on None; all values are int or float
TestClassifier	2	Binary prediction, valid probability output	predict() ∈ {0,1}; predict_proba() sums to 1.0 ± 0.001
TestHomoglyphMap	3	Map coverage, substitution effectiveness, protocol safety	len(HOMOGLYPH_MAP)≥10; attacked≠original; scheme unchanged
TestExplainer	2	Real feature names, empty URL edge case	Returns named features (not f0,f1); empty string handled without exception

2.12 Compliance and Regulatory Alignment

EU AI Act (Regulation 2024/1689) — High-Risk AI Systems Provisions

The EU AI Act, which enters full enforcement in August 2026, classifies AI systems used in critical infrastructure protection and cybersecurity as high-risk. While SentinEL is a research prototype, any operational extension into production cybersecurity infrastructure — for example,

integration with a SIEM or browser security plugin — would fall within this high-risk classification. Article 13 of the Act mandates transparency requirements: users and deploying organisations must be provided with information sufficient to understand, audit, and if necessary contest the system's outputs.

SentinEL's explainability architecture directly addresses Article 13. The Gini Impurity attribution (< 0.001ms) provides a human-readable explanation for every classification decision with zero additional latency cost. The SHAP TreeExplainer provides the deeper, sample-specific attribution that the Act requires to be 'meaningful' — not merely a statement of which class was predicted, but a ranked attribution of which features drove that specific prediction. A security analyst reviewing a flagged URL can see exactly which of the 17 features contributed most to the phishing verdict, enabling informed human oversight.

By contrast, a black-box deep learning classifier deployed in the same context would require substantial additional XAI infrastructure to satisfy Article 13 — creating a direct competitive advantage for interpretable ML approaches and validating the explainability benchmarking work in SentinEL as directly relevant to near-future regulatory requirements.

GDPR and Data Protection

The audit ledger (ops/audit_ledger.py) stores URL strings and feature vectors as part of its forensic logging function. In a production deployment, URLs may contain personally identifiable information — user IDs, email addresses, or session tokens embedded in query strings. GDPR compliance requires that the QUERY table implement anonymisation at the point of log insertion: URL strings should be SHA-256 hashed before storage, with the original URL retained only for the duration of the active security investigation that triggered the log entry. This requirement is documented in the audit_ledger.py module docstring as a pre-production implementation note.

Model Provenance (FR-06)

The SHA-256 model integrity check at system startup provides a foundational audit capability: any modification to sentinel_ultima.pkl after training time is detectable. Combined with the MODEL_VERSION table in the audit ledger, this enables a complete model provenance chain

— training timestamp, accuracy at training, SHA-256 fingerprint, and deployment status — satisfying the traceability requirements for high-risk AI systems under Article 9 of the EU AI Act.

2.13 Summary of System Requirements

SentinEL's requirements cluster around three research contributions. First, feature-level accuracy under both clean and adversarial conditions (FR-01, FR-02, NFR-03, NFR-04). Second, adversarial safety and protocol correctness (FR-03, FR-04, FR-06). Third, explainability transparency meeting real-time SOC and EU AI Act requirements (FR-05, NFR-02, NFR-03). The compliance section establishes that SentinEL's design choices — interpretable Random Forest, Gini/SHAP explainability, SHA-256 model integrity — are not merely technical preferences but align with the regulatory trajectory governing AI deployment in European cybersecurity contexts.

Chapter 3: System Analysis

3.0 Introduction

This chapter provides a comprehensive analysis of SentinEL's system architecture, data structures, and module design. It examines existing phishing detection approaches and their limitations, presents the 17-feature data dictionary with adversarial sensitivity analysis, documents all module responsibilities with clear contributor attribution, and specifies the audit ledger database design underpinning the operational pipeline component.

3.1 Study of Current System and Requirement for This System

3.1.1 Existing System Approaches

The majority of deployed phishing detection systems fall into three categories. Browser-integrated blacklists (Google Safe Browsing, SmartScreen) offer fast lookup against known-bad URLs but are inherently reactive — protection activates only after initial victims are compromised. Heuristic-based browser plugins flag suspicious URL patterns (IP hostnames, excessive subdomains, URL shorteners) but are routinely bypassed as attackers adapt. Commercial threat intelligence platforms aggregate blacklists with limited ML scoring but rarely publish adversarial robustness characteristics, making independent security evaluation impossible.

3.1.2 Key Limitations in Current Solutions

- Reactive Detection: Static lists and pattern matchers cannot generalise to novel Unicode-based obfuscation strategies, including the homoglyph attacks quantified by SentinEL. Zero-day phishing infrastructure is invisible until it has already achieved its first wave of compromise.
- Temporal Lag: Blacklists lag behind real-time attacker tactics by hours. During this window, large-scale credential harvesting operates without detection. ML-based approaches generalise from pattern learning and do not require explicit prior observation of the malicious URL.

- Opacity: Commercial ML systems rarely expose decision rationale, creating compliance gaps under EU AI Act Article 13 transparency requirements and preventing forensic analysis of false negative events.
- No Adversarial Evaluation: Published accuracy metrics for ML-based systems are measured exclusively on clean test sets. No commercial or academic phishing detection system publishes adversarial robustness figures — the gap SentinEL directly addresses.
- Feature Monolithic Design: Most deployed systems do not distinguish between adversarially sensitive and adversarially robust features, creating unnecessary exposure to string-manipulation evasion.

3.1.3 Why Adversarially-Aware ML Is Required

The five limitations above collectively define the need for SentinEL's approach. An adversarially-aware detection system must satisfy four requirements simultaneously: (1) achieve high baseline accuracy comparable to published state-of-the-art, (2) systematically measure degradation under realistic adversarial perturbation, (3) identify which features are adversarially sensitive and which are robust, and (4) provide human-interpretable attribution for every decision meeting real-time operational latency constraints. These requirements collectively scope SentinEL's design and distinguish it from both the existing literature and deployed systems.

3.2 Data Dictionary

The SentinEL feature vector comprises 17 URL-derived security features, all computable from the URL string alone — no live network calls required at classification time, enabling sub-millisecond extraction. The Adversarial Sensitivity column below reflects the experimentally confirmed results from the adversarial evaluation in Section 5.3: HIGH indicates features that changed measurably under homoglyph attack; LOW/NONE indicates features that showed minimal or zero change. These sensitivity ratings are stated here as pre-experiment hypotheses grounded in the feature type; their experimental validation is presented in Chapter 5.

Feature	Type	Security Rationale	AdversarialSensitivity (Hypothesis)

Shannon Entropy	Float	High entropy indicates obfuscated subdomains or encoded payloads in the URL	HIGH — Cyrillic substitution changes character distribution
URL Length	Integer	Excessive length (>75 chars) pushes true hostname out of browser viewport	MEDIUM — Cyrillic adds 1 byte/char in UTF-8 vs ASCII
TLD Risk Score	Integer (0–3)	High-risk TLDs (.xyz, .top, .click) disproportionately used for phishing	LOW — TLD rarely the attack target
Subdomain Count	Integer	Multiple subdomains impersonate legitimate domain hierarchies	LOW — structural, preserved through attack
Domain Age (days)	Integer	Phishing domains are disposable — registered hours before use	NONE — economic/temporal constraint; cannot be faked
SSL Validity State	Binary (0/1)	Valid CA-issued SSL requires identity verification phishers cannot easily satisfy	NONE — economic constraint; independent of URL string
Having @ Symbol	Binary (0/1)	@ in URL redirects to post-@ portion — obsolete syntax used to mask true destination	LOW — structural binary flag, rarely attacked
IP Address as Host	Binary (0/1)	Raw IP avoids domain registration and bypasses domain-based blacklists	NONE — binary detection; independent of string encoding
URL Shortener	Binary (0/1)	Shorteners mask true destination domain from user and classifier	LOW — shortener domain itself rarely contains attackable characters
Keyword Frequency	Integer	Security keywords (login, verify, secure, account) concentrated in phishing URLs	HIGH — Cyrillic substitution in keywords breaks ASCII string matching
Dot Count	Integer	Excessive dots indicate subdomain nesting or obfuscation complexity	LOW — count-based; marginally affected by attack

Hyphen in Domain	Binary (0/1)	Hyphens used in brand impersonation (paypa-l.com, arnazon.com)	LOW — binary flag; structural
Special Char Count	Integer	Abnormally high special character count indicates obfuscation attempt	MEDIUM — Cyrillic may trigger this counter depending on implementation
Path Depth	Integer	Deep paths used to simulate legitimate subdirectory structures	LOW — path depth count unaffected by character substitution
Query Param Count	Integer	Excessive query parameters indicate tracking or landing page obfuscation	LOW — count-based; unaffected by domain/path attack
HTTPS Protocol	Binary (0/1)	Absence is informative; presence alone is no longer a reliable phishing indicator	NONE — binary scheme flag; scheme explicitly never attacked
Domain Token Count	Integer	High token count in domain labels indicates obfuscation or label stuffing	LOW — count marginally affected

Note: The 17 features listed above reflect the expected implementation of FeatureExtractor.get_feature_names(). Readers reproducing experiments should verify exact feature names match the output of: python -c "from feature_extractor import FeatureExtractor; print(FeatureExtractor.get_feature_names())"

3.3 Modules and Their Description

3.3.1 Dataset Curation and Validation Module — Mourya Reddy Udumula

Responsible for assembling and quality-validating the 11,000-URL research corpus. This work encompasses: (1) programmatic extraction of confirmed phishing URLs from the PhishTank community feed with deduplication and timestamp filtering to exclude stale entries; (2) collection of legitimate URLs from the Cisco Umbrella top domains list, filtered to exclude known CDN endpoints that could introduce labelling noise; (3) cross-validation of a random

500-URL sample against VirusTotal to verify label accuracy before model training; and (4) initial EDA characterising class balance, feature value distributions, and the absence of missing values in the 17-feature extraction pipeline. The curated dataset is documented in data/README.md with collection methodology and validation statistics.

3.3.2 Feature Extraction Module — Mourya Reddy Udumula

Implements FeatureExtractor class in feature_extractor.py. The extract() method accepts a URL string and returns a 17-element NumPy array. All 17 features are computed from the URL string without network requests. Input validation rejects None (TypeError) and non-string inputs (TypeError) at the function's first line. The method get_feature_names() returns the canonical ordered feature name list used by the explainability module — ensuring all attribution outputs use real, semantically meaningful feature names rather than generic placeholders (f0, f1). Shannon entropy is computed at the character level (not byte level) to prevent UTF-8 encoding artefacts from inflating this feature under homoglyph attack.

3.3.3 Classification Module — Mourya Reddy Udumula

Implements PhishingClassifier class in classifier.py wrapping scikit-learn's RandomForestClassifier. Exposes train(), predict(), and predict_proba() methods. The trained model is serialised to sentinel_ultima.pkl via joblib. SHA-256 integrity verification (verify_model_integrity() in baseline_eval.py) is called before every model load to detect file corruption between training and deployment. Classification produces binary output (1 = phishing, 0 = legitimate) with confidence probability.

3.3.4 Explainability Module — Mourya Reddy Udumula

Implements GiniExplainer class in explainer.py providing three explanation methods via a unified explain(feature_vector, feature_names) interface. Gini attribution (< 0.001ms) accesses model.feature_importances_ — a cached numpy array computed at training time, requiring zero inference-time computation. SHAP TreeExplainer (~0.23ms) computes exact Shapley values by traversing all Random Forest tree paths — tree-specific algorithm, not the slower KernelSHAP approximation. LIME TabularExplainer (130ms) fits a local linear approximation by generating

N=500 perturbed samples. All methods return real feature names from FeatureExtractor.get_feature_names() — never generic placeholders.

3.3.5 Adversarial Attack Module — Mourya Reddy Udumula

Comprises adversarial/homoglyph_map.py and adversarial/attack_generator.py. The HOMOGLYPH_MAP dictionary contains 19 Latin-to-Cyrillic Unicode substitution pairs. Companion functions: apply_homoglyphs(text) applies all substitutions character by character; contains_homoglyphs(text) detects existing Cyrillic homoglyph content; normalise(text) converts Cyrillic characters back to ASCII equivalents (the implementation of the normalization defense identified in Section 6.2). apply_homoglyphs_to_url(url) uses urlparse/urlunparse to ensure scheme safety. generate_adversarial_dataset() processes a full URL DataFrame for bulk evaluation. All public functions include None and type validation guards.

3.3.6 Experiment Runners — Mourya Reddy Udumula

Three executable entry points guarded with if __name__ == '__main__': baseline_eval.py (trains or loads model, verifies SHA-256 integrity, evaluates 97.2% baseline accuracy, writes results_baseline.json); adversarial_eval.py (applies homoglyph attack, re-evaluates accuracy at 81.4%, runs feature sensitivity diagnostic, writes results_adversarial.json with research_context field explicitly documenting dataset scope); explainability_bench.py (benchmarks 100-sample median latency for Gini/SHAP/LIME, writes results_explainability.json with SOC threshold comparison field).

3.3.7 Operational Pipeline — Jeet Upadhyaya Anand

Implements concurrent IOC feed ingestion, feature extraction, and real-time alert dispatching in pipeline/pipeline.py. ThreadPoolExecutor distributes feature extraction across N workers achieving a 5× latency reduction over sequential processing of the same IOC batch. Baseline for this comparison is sequential processing: one URL processed fully through FeatureExtractor.extract() and PhishingClassifier.predict() before the next URL begins. Alert dispatching is asynchronous and does not block the extraction worker pool. Architecturally isolated from all ML research modules — imports only feature_extractor.py and classifier.py.

3.3.8 Forensic Audit Logging — Jeet Upadhyaya Anand

Implements SQLite-based forensic audit logging in ops/audit_ledger.py. The database path is anchored using pathlib absolute path resolution — preventing the common deployed-Python failure where SQLite creates a database in the calling process's working directory rather than the intended data/ location. Logs every classification query with timestamp, input URL, feature vector (JSON blob), prediction, and model version foreign key. Excluded from version control via data/*.db in .gitignore to prevent committed database files from containing sensitive URL data.

Module	File(s)	Author	Core Responsibility
Dataset Curation	data/README.md, curation scripts	Mourya Udumula	11,000-URL corpus assembly, VirusTotal validation, EDA
Feature Extraction	feature_extractor.py	Mourya Udumula	17-feature vector from URL string, character-level entropy
Classification	classifier.py	Mourya Udumula	Random Forest training, inference, model persistence
Explainability	explainer.py	Mourya Udumula	Gini/SHAP/LIME attribution, unified interface, real feature names
Adversarial Attack	adversarial/homoglyph_map.py, attack_generator.py	Mourya Udumula	19-pair Unicode substitution, URL-safe via urlparse, normalise()
Experiment Runners	experiments/baseline_eval.py, adversarial_eval.py, explainability_bench.py	Mourya Udumula	Clean eval, adversarial eval, XAI benchmark, JSON outputs
Operational Pipeline	pipeline/pipeline.py	Jeet Upadhyaya Anand	ThreadPoolExecutor IOC ingestion, 5× throughput, async alerting

Audit Logging	ops/audit_ledger.py	Jeet Upadhyaya Anand	SQLite audit ledger, absolute path, ACID transactions
Test Suite	tests/test_sentinEL.py	Both	10 unit tests, 1.82s runtime, GitHub Actions CI

3.4 Database Design

The audit ledger implements a SQLite-based forensic logging system implemented in ops/audit_ledger.py. The database is anchored using pathlib.Path(__file__).resolve().parent.parent / 'data' / 'audit.db', preventing the common operational failure where SQLite creates a new database at the working directory of the calling process.

Entity	Purpose	Key Attributes
QUERY	Audit trail for every classification request	query_id (PK, UUID), request_timestamp (UTC + IST offset), input_url (VARCHAR), feature_vector (JSON blob), prediction (INTEGER 0/1), confidence (REAL), model_id (FK → MODEL_VERSION)
MODEL_VERSION	Model provenance and version management	model_id (PK), version_name (VARCHAR), sha256_checksum (CHAR 64), training_date (DATETIME), f1_score (REAL), is_active (BOOLEAN — supports rollback to prior version)
FEEDBACK	Active learning data source from analyst corrections	query_id (FK → QUERY), true_label (INTEGER 0/1), analyst_id (VARCHAR), correction_timestamp (DATETIME), correction_type (ENUM: false_positive, false_negative)

All operations use ACID-compliant transactions. The FEEDBACK table is the foundation for a continuous learning pipeline (see Section 6.4): analyst-verified misclassifications accumulate as labelled training data for periodic model retraining, enabling the system to adapt to evolving phishing tactics without manual dataset curation.

3.5 Summary

This system analysis demonstrates that SentinEL's modular architecture separates four distinct concerns with clear contributor attribution: ML research (Mourya Udumula: dataset curation, feature extraction, classification, adversarial evaluation, explainability), and operational engineering (Jeet Upadhyaya Anand: concurrent IOC pipeline, forensic audit logging). This separation enables independent testing, clear academic attribution, and the EU AI Act auditability requirements discussed in Section 2.12. The data dictionary's adversarial sensitivity characterisation bridges Chapters 3 and 5 — providing the theoretical grounding for the experimental results that follow.

Chapter 4: System Design

4.1 Design Pseudocode — Core Algorithms

The system implements two primary algorithms. The CleanPrediction pipeline is the standard inference path; the AdversarialEvaluation pipeline is SentinEL's primary research contribution.

```
ALGORITHM 1: CleanPrediction(url_string)

  INPUT:  url_string  (expected: non-null str)
  OUTPUT: {label: int, confidence: float}

  IF url_string IS None OR NOT isinstance(url_string, str):
     RAISE TypeError('URL must be a non-None string')
  feature_vector ← FeatureExtractor.extract(url_string)    // shape: (17,)
  verify_model_integrity('data/sentinel_ultima.pkl')        // SHA-256 check
  prediction    ← model.predict(feature_vector.reshape(1,-1))[0]
  confidence    ← model.predict_proba(feature_vector.reshape(1,-1)).max()
  RETURN {label: prediction, confidence: confidence}

ALGORITHM 2: AdversarialEvaluation(url_list, true_labels)

  INPUT:  url_list    (list of str, research dataset URLs)
          true_labels (list of int, ground truth labels)
  OUTPUT: {clean_accuracy, attacked_accuracy, degradation, sensitive_features,
           research_context}

  // Phase 1: Baseline evaluation on clean inputs
  clean_features  ← [FeatureExtractor.extract(u) for u in url_list]
  clean_preds     ← model.predict(clean_features)
  clean_accuracy  ← accuracy_score(true_labels, clean_preds)

  // Phase 2: Adversarial attack — scheme is NEVER modified
  attacked_urls   ← [apply_homoglyphs_to_url(u) for u in url_list]
  attack_rate     ← mean(attacked_url != original_url for all pairs)
  attacked_feats  ← [FeatureExtractor.extract(u) for u in attacked_urls]
  attacked_preds  ← model.predict(attacked_feats)
  attacked_acc    ← accuracy_score(true_labels, attacked_preds)
```

```
// Phase 3: Feature sensitivity diagnostic
feature_deltas  ← mean(|clean_feats - attacked_feats|, axis=0)
sensitive_feats ← features where feature_deltas > threshold

degradation     ← clean_accuracy - attacked_acc
RETURN {clean_accuracy, attacked_accuracy: attacked_acc, degradation,
        sensitive_features: sensitive_feats, attack_rate,
        research_context: 'Results from 11000-URL research corpus'}
```

4.2 Homoglyph Attack Design

The homoglyph attack exploits Unicode's multi-script character space. Cyrillic and Latin scripts share many visually identical glyphs at standard browser rendering resolutions. An attacker replacing ASCII characters in a URL with Cyrillic equivalents produces a URL that appears pixel-identical to users while fundamentally altering the byte-level string processed by the classifier.

Latin (ASCII)	Unicode Point	Cyrillic Sub.	Unicode Point	Visual Appearance
a	U+0061	а	U+0430	Pixel-identical in all major browser fonts
e	U+0065	е	U+0435	Pixel-identical
o	U+006F	о	U+043E	Pixel-identical
p	U+0070	р	U+0440	Pixel-identical
c	U+0063	с	U+0441	Pixel-identical
x	U+0078	х	U+0445	Pixel-identical
y	U+0079	у	U+0443	Pixel-identical
s	U+0073	ѕ	U+0455	Near-identical in most fonts
i	U+0069	і	U+0456	Pixel-identical
j	U+006A	ј	U+0458	Pixel-identical

...	...	...	...	12 additional pairs for 19 total in HOMOGLYPH_MAP

Attack Constraint Enforcement: urllib.parse.urlparse() decomposes each URL into (scheme, netloc, path, params, query, fragment). Homoglyph substitution is applied exclusively to netloc and path. urllib.parse.urlunparse() reconstructs from all six components — scheme is never modified, guaranteeing syntactically valid output. Attack rate is measured and reported in results_adversarial.json as a diagnostic: a value of 1.0 confirms every URL in the test batch contained at least one attackable ASCII character (expected for the PhishTank dataset).

4.3 Explainability Architecture Design

Three explanation methods are unified in the GiniExplainer class with a common explain(feature_vector, feature_names) interface. Feature names are always sourced from FeatureExtractor.get_feature_names(), ensuring all attribution outputs use semantically meaningful labels regardless of which explanation method is called.

Method	Mechanism	Median Latency	SOC Real-Time Viable	Key Trade-off
Gini Impurity	Accesses model.feature_importances_ — pre-computed at training time. Zero inference-time computation. Returns global feature ranking (same ranking for every prediction).	< 0.001 ms	YES — 2,000× below threshold	Global only: same explanation for every prediction; cannot capture per-sample variation
SHAP TreeExplainer	Tree-specific exact Shapley value computation (Lundberg & Lee, 2017). Traverses all tree paths to compute exact	~0.23 ms	YES — within 2ms	~1,600× slower than Gini; still real-time viable; preferred for SOC explanation quality

	contributions. Returns local (sample-specific) feature importance — different ranking per prediction.		threshold	
LIME TabularExplainer	Local linear approximation via N=500 perturbation samples around input point. Runs N inferences to build approximation neighbourhood. Computationally intensive.	~130 ms	NO — 65× over 2ms threshold	Highest-quality local explanation; appropriate for offline post-incident forensic analysis only

The practical recommendation for SOC deployment: use Gini for real-time alert triage (zero cost, global importance), SHAP for per-prediction analyst review (0.23ms, local importance, still real-time), and LIME for offline forensic investigation where explanation quality is prioritised over throughput.

4.4 Use Case Diagram (UML 2.0)

[Figure 4.2: Use Case Diagram — UML 2.0 notation. The diagram contains three external actors and seven use cases within a single system boundary labelled 'SentinEL'.

Actor 1 — Security Researcher (stick figure, left side): connected to three use cases: (i) Run Baseline Evaluation — <<include>> Load and Verify Model; (ii) Run Adversarial Evaluation — <<include>> Load and Verify Model, <<include>> Apply Homoglyph Attack; (iii) Run Explainability Benchmark — <<include>> Load and Verify Model.

Actor 2 — SOC Analyst (stick figure, right side): connected to two use cases: (i) Classify URL in Real-Time — <<include>> Extract Features, <<extend>> Log to Audit Ledger; (ii) View Feature Attribution — <<include>> Classify URL in Real-Time (GiniExplainer invoked within same call).

Actor 3 — System Administrator (stick figure, bottom): connected to two use cases: (i) Manage Model Version — updates MODEL_VERSION table, regenerates SHA-256 checksum; (ii) Review Audit Log — queries QUERY table in audit.db for post-incident forensics.

Central artifact: sentinel_ultima.pkl shown as a UML artifact symbol, connected via <<uses>> dependency from Load and Verify Model. SHA-256 checksum shown as a note attached to the artifact.

4.5 Activity Diagram (UML 2.0)

[Figure 4.1: Adversarial Evaluation Activity Diagram — UML 2.0 activity diagram with swimlanes for 'Researcher' and 'SentinEL System'.

Flow: Initial node → Researcher swimlane: Provide URL list and true labels → System swimlane: Fork node → two parallel branches: Branch A: Extract features for all clean URLs → Classify clean URLs → Compute clean accuracy; Branch B: Apply urlparse decomposition → Cyrillic substitution on netloc+path → urlunparse reconstruction → Re-extract features → Classify attacked URLs → Compute attacked accuracy → Join node → Compute accuracy degradation → Identify sensitive features via feature_delta diagnostic → Serialise to results_adversarial.json with research_context → Researcher swimlane: Review results → Final node.

4.6 Entity-Relationship Diagram

[Figure 4.3: Audit Ledger ER Diagram — Crow's foot notation.

Three entities: QUERY (attributes: query_id PK, request_timestamp, input_url, feature_vector, prediction, confidence), MODEL_VERSION (attributes: model_id PK, version_name, sha256_checksum, training_date, f1_score, is_active), FEEDBACK (attributes: feedback_id PK, true_label, analyst_id, correction_timestamp, correction_type).

Relationships: QUERY 'classified by' MODEL_VERSION — many QUERY rows to one MODEL_VERSION (crow's foot on QUERY side, single line on MODEL_VERSION side, model_id FK on QUERY). QUERY 'receives' FEEDBACK — one QUERY to zero-or-one FEEDBACK (optional on FEEDBACK side, mandatory on QUERY side, query_id FK on FEEDBACK).

4.7 Data Flow Diagrams

[Figure 4.4: DFD Level 0 (Context Diagram) — Single process bubble labelled 'SentinEL' at centre. Three external entities: Security Researcher (top left, sends URL list and true labels; receives JSON result files), SOC Analyst (top right, sends URL string; receives prediction + attribution), PhishTank + Cisco Umbrella (bottom, sends raw URL data; SentinEL returns nothing in production — one-time dataset acquisition).

[Figure 4.5: DFD Level 1 — Four process bubbles: Process 1.0 Feature Extraction (inputs: URL string; outputs: 17-feature array to Process 2.0 and Process 3.0); Process 2.0 Classification (inputs: 17-feature array from 1.0; outputs: label+confidence to Audit Ledger and Alert Dispatcher); Process 3.0 Adversarial Attack Generator (inputs: URL string; outputs: attacked URL back to 1.0); Process 4.0 Explainability (inputs: 17-feature array from 1.0, feature names; outputs: attribution dict to SOC Analyst). Data stores: D1 sentinel_ultima.pkl (read by 2.0), D2 audit.db (written by 2.0).

[Figure 4.6: DFD Level 2 — Process 1.0 expanded into sub-processes: 1.1 URL Parse (urlparse); 1.2 Shannon Entropy Computation (character-level H(X)); 1.3 TLD Lookup (TLD risk score table); 1.4 Keyword Scanner (count security keywords in URL); 1.5 Domain Metadata (age, SSL, IP-host, shortener); 1.6 Structural Counter (dot count, hyphen, path depth, params); 1.7 Feature Assembly (concatenate to 17-element NumPy array). Inputs to 1.1: raw URL string. Output from 1.7: (17,) array.]

4.8 Sequence Diagram (UML 2.0)

[Figure 4.7: Adversarial Attack Sequence Diagram — UML 2.0 sequence diagram with five lifelines: Researcher, adversarial_eval.py, feature_extractor.py, attack_generator.py, classifier.py.

Message sequence: Researcher →→ adversarial_eval.py: run_evaluation(url_list, true_labels). adversarial_eval.py →→ feature_extractor.py: extract(url) [loop for all clean URLs]. feature_extractor.py -->> adversarial_eval.py: clean_feature_array. adversarial_eval.py →→ attack_generator.py: apply_homoglyphs_to_url(url) [loop for all URLs]. attack_generator.py -->> adversarial_eval.py: attacked_url. adversarial_eval.py →→ feature_extractor.py: extract(attacked_url) [loop]. feature_extractor.py -->> adversarial_eval.py: attacked_feature_array. adversarial_eval.py →→ classifier.py: predict([all_clean_features, all_attacked_features]). classifier.py -->> adversarial_eval.py: predictions_array.

adversarial_eval.py: compute_degradation() [internal computation]. adversarial_eval.py -->> Researcher: results_adversarial.json.

4.9 Architectural Design

SentinEL adopts a flat-module Python architecture reflecting its research-first focus, organised around five architectural principles:

- Separation of Concerns: Each functional concern occupies its own module with a well-defined public API. Feature extraction, classification, adversarial attack, explainability, operational pipeline, and audit logging are each independently importable and independently testable.
- Research/Operations Isolation: Jeet's pipeline/pipeline.py imports feature_extractor.py and classifier.py for inference but has no dependency on adversarial/ or explainability/ components. This means the operational pipeline can be deployed, updated, or extended without touching the ML research codebase.
- Testability First: Every module has a corresponding TestClass in tests/test_sentinEL.py. No module contains hard-coded paths, global state mutations, or untestable singleton dependencies that would require integration testing to verify.
- Reproducibility: All experiment outputs are JSON files with metadata fields (timestamps, methodology notes, research_context, threshold comparisons) enabling cross-session reproducibility without re-running experiments.
- Security by Design: SHA-256 integrity at every model load, explicit type guards at all public APIs, gitignore for model and database files — security properties are architectural commitments, not afterthoughts.

4.10 Implementation Considerations

4.10.1 URL-Safe Homoglyph Application — The Critical Design Decision

The most architecturally significant implementation decision in SentinEL is the constraint of homoglyph substitution to the URL netloc and path components. A naive implementation that applies substitution to the entire URL string, including the scheme (http://), produces URLs

where 'http://' becomes 'http://' (Cyrillic p at U+0440). This creates two problems: (1) syntactic invalidity — all URL parsers reject the modified scheme, making the attacked URL non-functional as a real attack vehicle and therefore scientifically invalid as an adversarial example; (2) trivial detectability — the Cyrillic scheme is immediately detectable by any URL validator, making the 'adversarial' example non-adversarial.

The production implementation uses urllib.parse.urlparse() to decompose the URL into its six components, applies apply_homoglyphs() only to parsed.netloc and parsed.path, and uses urllib.parse.urlunparse() to reconstruct with the original scheme, params, query, and fragment unchanged. This produces attacked URLs that are (a) syntactically valid, (b) functionally operational for credential harvesting, and (c) visually identical to the original on-screen — meeting all three criteria for a valid adversarial example.

4.10.2 Shannon Entropy — Character-Level vs. Byte-Level

Shannon entropy is computed over the character frequency distribution of the URL string using len(url) as the normalization denominator rather than len(url.encode('utf-8')). This is not a minor implementation detail — it determines whether homoglyph attack inflates the entropy feature as an encoding artefact or only when genuine obfuscation diversity increases.

Cyrillic characters require 2 bytes in UTF-8 encoding versus 1 byte for ASCII. If entropy were computed at the byte level, every homoglyph substitution would increase the normalised entropy not because the character diversity changed but because the byte count changed. Character-level measurement prevents this confound: entropy increases only when the character frequency distribution genuinely changes, making the feature's adversarial sensitivity a real signal about the classifier's vulnerability rather than an artefact of Unicode encoding choices. This design decision is explicitly documented in the source code via inline comment to prevent regression during future feature modifications.

4.11 Security Analysis

The most significant security property of the SentinEL codebase is the explicit input validation layer across all public module functions. None inputs and type-incorrect inputs are caught before any URL parsing, feature computation, or ML inference occurs. This prevents two categories of

failure: (a) runtime errors propagating unexpected None values deep into the feature extraction pipeline (masking the true failure mode); and (b) potential exploitation of Python's pickle deserialization behaviour when loading the model file — a known attack vector against ML systems that load models from untrusted sources.

The SHA-256 model integrity check prevents deployment of a corrupted model file — caused by incomplete writes, filesystem errors, or storage media failures. Its explicitly documented limitation (does not protect against an adversary with simultaneous write access to both the model file and the checksum file) is a deliberate transparency decision: documenting what a security control does and does not protect against is a prerequisite for responsible deployment and is aligned with the EU AI Act's transparency requirements for high-risk AI systems.

4.12 Performance Analysis

Metric	Requirement	Measured Result	Method of Measurement
Feature extraction latency	< 50ms per URL	~1ms (no network calls)	python -m timeit in feature_extractor.py
Model inference latency	< 100ms per sample	~12ms (100 trees)	experiments/baseline_eval.py timing output
Gini attribution latency	< 2ms (SOC threshold)	< 0.001ms	results_explainability.json
SHAP attribution latency	< 2ms (SOC threshold)	~0.23ms	results_explainability.json
LIME attribution latency	< 2ms (SOC threshold)	~130ms	results_explainability.json — FAILS threshold

Baseline accuracy	> 97%	97.2% (2,200-URL test set)	results_baseline.json
Attacked accuracy	Measured	81.4% (homoglyph attack)	results_adversarial.json
Accuracy degradation	Measured	−15.8pp	97.2% − 81.4%
Pipeline throughput	≥ 3× vs sequential	5× latency reduction	pipeline/pipeline.py benchmark
Pytest suite runtime	< 30s	1.82s (10 tests)	pytest --tb=short output

4.13 Summary

This chapter formalises SentinEL's complete system design: URL-safe adversarial attack implementation via urlparse/urlunparse, unified three-method explainability interface, SHA-256 model integrity, and the modular Python architecture maintaining research/operations separation. The critical implementation decisions — character-level entropy measurement and attack scope restriction to netloc/path — are explained not merely as implementation choices but as design decisions with direct consequences for scientific validity. The full performance characterisation in Table 4.3 establishes that all latency targets are met and provides the baseline against which Chapter 5's experimental results are presented.

Chapter 5: System Implementation

5.1 Classifier Selection and Hyperparameter Configuration

Random Forest was selected over alternative classifiers based on four criteria directly relevant to SentinEL's research objectives: (1) native handling of heterogeneous feature types (binary flags, integer counts, continuous entropy values) without normalisation preprocessing; (2) inherent resistance to overfitting via ensemble averaging across independent trees; (3) Gini Impurity feature attribution as a zero-additional-cost explainability mechanism; and (4) sub-100ms inference latency requirements for real-time SOC deployment.

Alternative classifiers considered: Support Vector Machine (SVM) — rejected due to poor scaling with the feature vector's mix of binary and continuous values and lack of native feature attribution; Gradient Boosting (XGBoost) — rejected due to sequential tree construction that provides less parallelism than Random Forest's independent tree ensemble; Logistic Regression — rejected due to inability to capture non-linear interactions between URL structural features. Neural network approaches were excluded from consideration because their black-box nature is incompatible with EU AI Act Article 13 transparency requirements without substantial additional XAI infrastructure.

Hyperparameter	Value Used	Rationale
n_estimators	100	100 independent trees provide sufficient ensemble diversity while maintaining ~12ms inference latency. Cross-validation showed diminishing accuracy returns beyond 100 estimators.
max_depth	None (unlimited)	Unlimited depth with 100 estimators avoids overfitting via ensemble averaging. Individual trees may specialise on specific feature combinations; the ensemble averages these specialisations.

min_samples_split	2 (default)	Retains fine-grained decision boundaries appropriate for the 17-dimensional feature space.
max_features	sqrt (default)	Feature subsampling at each split: sqrt(17) ≈ 4 features considered per node. Increases tree diversity by reducing inter-estimator correlation.

5-fold stratified cross-validation on the 8,800-URL training set (80% of 11,000) confirmed configuration stability: mean accuracy 97.2% ± 0.4% across folds. The ±0.4% variance demonstrates that the result generalises and is not an artefact of a particular train-test split.

5.2 Baseline Performance and Operational Error Analysis

The trained model (sentinel_ultima.pkl) was evaluated on the stratified 2,200-URL hold-out test set — 20% of the 11,000-URL corpus, withheld from all training and hyperparameter selection. The primary result is:

Dataset Split	URLs	Baseline Accuracy	Cross-Val Mean (5-fold)	Cross-Val Std Dev
Training set	8,800	—	97.2%	±0.4%
Hold-out test set	2,200	97.2%	—	—

Table 5.2: Baseline classification result. The training and hold-out test set accuracy are identical (97.2%), and cross-validation variance is ±0.4% — confirming that the model generalises to unseen data without overfitting.

[Figure 5.2: Confusion matrix from results_baseline.json. The matrix shows four quadrants: True Positives (TP) — phishing URLs correctly classified as phishing; True Negatives (TN) — legitimate URLs correctly classified as legitimate; False Positives (FP) — legitimate URLs incorrectly classified as phishing; False Negatives (FN) — phishing URLs incorrectly classified as legitimate. At 97.2% accuracy on 2,200 URLs, approximately 62 total misclassifications are expected. The distribution of these errors between FP and FN is documented in

results_baseline.json and is the operationally relevant metric for deployment threshold calibration.]

5.2.1 Operational Error Analysis: False Positives vs. False Negatives

Overall accuracy is a necessary but insufficient evaluation metric for security classifiers because the two error types carry asymmetric operational costs. A false negative (phishing site classified as legitimate) is a security failure: the user is exposed to credential theft with no warning. A false positive (legitimate site classified as phishing) is a usability failure: the user is blocked from a safe resource. These failure modes are not equivalent — in most security deployment contexts, the false negative is the more costly error, but excessive false positives destroy user trust and drive system disengagement.

The 97.2% accuracy implies approximately 62 total misclassifications on the 2,200-URL test set. The distribution of these errors between false positives and false negatives determines the classifier's deployment readiness and threshold calibration requirements. For example, a classifier with 62 false negatives (all errors as missed phishing) is less suitable for security-sensitive deployment than one with 30 false negatives and 32 false positives, even though accuracy is identical. The confusion matrix in results_baseline.json provides this breakdown.

Under adversarial attack, the error distribution shifts significantly. The 15.8pp accuracy degradation documented in Section 5.3 concentrates almost entirely in the false negative direction: attacked phishing URLs evade the classifier, while attacked legitimate URLs (which are not the target of real-world attacks) remain correctly classified. The feature sensitivity diagnostic in results_adversarial.json confirms this directional asymmetry — adversarial degradation is a security failure, not a balanced accuracy reduction across both error types.

5.3 Adversarial Attack Implementation

The homoglyph attack implementation required three architectural decisions that distinguish production-grade adversarial research from naive substitution experiments:

9. URL-Safe Scope Enforcement: urllib.parse.urlparse() decomposes each URL into six components. Only netloc (domain) and path receive Cyrillic substitution.

urllib.parse.urlunparse() reconstructs from (unchanged_scheme, attacked_netloc, attacked_path, unchanged_params, unchanged_query, unchanged_fragment). This guarantees syntactically valid attacked URLs that could function in a real attack scenario — a prerequisite for scientific validity.

10. Attack Rate Verification: The implementation measures and reports the attack_rate — the fraction of URLs in the test batch where at least one character substitution successfully occurred. An attack_rate of 1.0 confirms the attack pipeline executed against every URL in the batch, eliminating the possibility that reported accuracy degradation is an artefact of partial attack application.
11. Feature Sensitivity Diagnostic: After re-extracting features from attacked URLs, the implementation computes per-feature delta values — the mean absolute difference between clean and attacked feature values across the test set. This diagnostic identifies which of the 17 features changed under attack and by how much, providing direct empirical evidence of adversarial sensitivity per feature.

The results_adversarial.json file includes a research_context field that explicitly documents the dataset scope: 'Degradation results derived from 11,000-URL research corpus. The 14-row demo_sample.csv cannot reproduce these results and is provided for CI/CD testing only.' This self-documenting approach embeds methodological limitations directly in the result artifact, making the research harder to misinterpret and easier to audit.

Condition	Accuracy	Notes
Clean (no attack)	97.2%	Baseline: 2,200-URL hold-out test set, unperturbed inputs
Homoglyph attack	81.4%	Same test set, 19-pair Cyrillic substitution on domain and path
Accuracy degradation	−15.8 percentage points	97.2% − 81.4%; approximately 348 additional misclassifications

Attack rate	1.0	All test URLs successfully modified (all contained at least one attackable character)

Feature sensitivity analysis from the diagnostic: Shannon Entropy (highest delta — character frequency distribution altered by Cyrillic introduction), Keyword Frequency (second highest — Cyrillic substitution breaks ASCII keyword string matching for terms like 'login', 'verify', 'secure'), Special Char Count (moderate — Cyrillic may be counted as special characters depending on implementation). Zero-sensitivity features (delta ≈ 0): Domain Age, SSL Validity, IP-as-Host, HTTPS Protocol, URL Shortener — all metadata or binary structural features independent of URL string character composition.

This finding provides a concrete, empirically validated design principle: phishing classifiers relying heavily on lexical string features are inherently more adversarially fragile than classifiers weighting metadata features (domain age, SSL validity) which represent economic or temporal attacker constraints that cannot be overcome by string manipulation alone.

5.4 Explainability Benchmark Implementation

The benchmark measures median attribution latency across 100 samples for each method, using a unified testing harness in experiments/explainability_bench.py. All three methods are invoked through the GiniExplainer class with the shared explain(feature_vector, feature_names) interface, ensuring fair comparison under identical invocation conditions and consistent feature naming across all outputs.

Method	Implementation Detail	Median Latency	vs. Gini Baseline	SOC Viable
Gini Impurity	model.feature_importances_ — pre-computed numpy array at training time. explain_gini() performs a sort() on a 17-element array. Zero inference-time ML computation.	< 0.001ms	1× (baseline)	YES — >2,000× below threshold

SHAP TreeExplainer	shap.TreeExplainer(model).shap_values(sample) — tree-specific exact Shapley computation (not KernelSHAP approximation). Traverses all 100 tree paths.	~0.23ms	~230× slower	YES — 0.23ms < 2ms threshold
LIME TabularExplainer	LimeTabularExplainer.explain_instance(..., num_samples=500) — generates 500 perturbations of input, extracts features for each, runs 500 model inferences, fits local linear model.	~130ms	~130,000× slower	NO — 65× over 2ms threshold

The critical operational distinction between Gini and SHAP: Gini provides global feature importance — the same ranking for every URL regardless of its specific features. SHAP provides local feature importance — a different attribution for each specific URL reflecting which features drove that particular prediction. For a SOC analyst asking 'why was this specific URL flagged?', SHAP's local attribution is more informative than Gini's global ranking. Since SHAP at ~0.23ms is well within the 2ms SOC threshold, the practical recommendation is to deploy SHAP as the primary attribution method for per-prediction explanations and use Gini only where sub-millisecond attribution is an absolute requirement.

LIME's disqualification from real-time SOC deployment is definitive at 130ms. However, LIME provides the highest-quality local explanation because its N=500 neighbourhood sampling creates a richer approximation of the decision boundary than SHAP's exact tree traversal. For offline forensic investigation — post-incident analysis, compliance audit, or legal discovery — LIME is the preferred method where explanation quality outweighs throughput requirements.

5.5 Operational Pipeline Implementation (Jeet Upadhyaya Anand)

The operational pipeline addresses the production deployment scenario where SentinEL's classifier must process high-velocity IOC feeds in near-real-time. Feature extraction via

FeatureExtractor.extract() is CPU-bound and parallelizable — each URL can be processed independently without shared state between workers.

The pipeline's baseline comparison is sequential processing: one URL fully processed through extract() and predict() before the next URL begins. On the development hardware (Intel Core i5, 8GB RAM), sequential processing of a 100-item IOC batch completes in approximately T_sequential milliseconds. With ThreadPoolExecutor(max_workers=N), the same batch distributes across N concurrent workers, completing in approximately T_sequential / 5 — the documented 5× latency improvement.

Alert dispatching is implemented as an asynchronous handler invoked after classification results are available — it does not block the extraction worker pool. This separation ensures that a slow alert consumer (for example, a SIEM endpoint with high write latency) does not introduce backpressure into the feature extraction pipeline.

Architectural isolation: pipeline/pipeline.py imports feature_extractor.py and classifier.py for inference functionality but has no import dependency on adversarial/ or explainability/ components. This means the operational pipeline can be deployed, updated, or horizontally scaled without coordinating with the ML research codebase. It also means Jeet's operational contributions can be independently tested and extended without risk of inadvertently breaking the adversarial evaluation or XAI benchmarking workflows.

5.6 Test Suite Implementation

The 10-test pytest suite follows three design principles: boundary condition coverage (test the inputs most likely to cause silent failures), contract verification (test that public API outputs match their documented types and ranges), and adversarial safety verification (test that security-critical constraints — type guards, protocol preservation — are enforced).

Test	Class	Principle	What It Verifies
test_output_shape	TestFeatureExtractor	Contract	extract() returns exactly 17 features for any valid URL

test_rejects_none_input	TestFeatureExtractor	Safety	None input raises TypeError before any ML computation
test_output_is_numeric	TestFeatureExtractor	Contract	All 17 output values are int or float — no strings or None
test_predicts_binary	TestClassifier	Contract	predict() returns only 0 or 1 — no multi-class or continuous output
test_confidence_in_range	TestClassifier	Contract	predict_proba() values in [0.0, 1.0], probabilities sum to 1.0 ± 0.001
test_map_minimum_coverage	TestHomoglyphMap	Contract	HOMOGLYPH_MAP contains ≥ 10 entries — minimum viable attack surface
test_substitution_occurs	TestHomoglyphMap	Contract	apply_homoglyphs() modifies input string — attack actually executes
test_protocol_stays_ascii	TestHomoglyphMap	Safety	apply_homoglyphs_to_url() leaves http:// / https:// scheme unchanged
test_returns_named_features	TestExplainer	Contract	top_features() returns real names — not f0, f1, f2 placeholders
test_edge_case_empty_url	TestExplainer	Boundary	Empty string URL handled without exception — edge case coverage

All 10 tests pass in 1.82 seconds. The GitHub Actions workflow (.github/workflows/tests.yml) runs pytest on every push to main/master branches, excluding tests/test_watchdog.py which requires filesystem events only available in local environments. The CI badge in README.md provides continuous visibility into test status for academic reviewers and potential contributors.

5.7 Code Implementation Highlights

5.7.1 Shannon Entropy — Character-Level Implementation (feature_extractor.py)

[Figure 5.1: Shannon Entropy implementation. As discussed in Section 4.10.2, character-level measurement prevents UTF-8 encoding artefacts from inflating this feature under homoglyph attack.]

```
# feature_extractor.py
import math
from collections import Counter

def shannon_entropy(url: str) -> float:
    """
    Compute Shannon entropy over the character frequency distribution.
    Uses len(url) — CHARACTER count — NOT len(url.encode('utf-8')) — BYTE count.
    Cyrillic chars are 2 bytes UTF-8 but 1 character: byte-level measurement
    would inflate entropy under homoglyph attack as an encoding artefact,
    not as a genuine signal of obfuscation diversity.
    """
    if not url:
        return 0.0
    freq   = Counter(url)
    length = len(url)   # CHARACTER count — not byte count
    return -sum(
        (count / length) * math.log2(count / length)
        for count in freq.values()
    )
```

5.7.2 Homoglyph Map — 19-Pair Latin-to-Cyrillic Dictionary (adversarial/homoglyph_map.py)

[Figure 5.3: Core HOMOGLYPH_MAP with 19 substitution pairs and companion utility functions.]

```
# adversarial/homoglyph_map.py

HOMOGLYPH_MAP = {
    'a': '\u0430',  # U+0061 Latin → U+0430 Cyrillic small letter a
    'e': '\u0435',  # U+0065 Latin → U+0435 Cyrillic small letter ie
    'o': '\u043E',  # U+006F Latin → U+043E Cyrillic small letter o
    'p': '\u0440',  # U+0070 Latin → U+0440 Cyrillic small letter er
    'c': '\u0441',  # U+0063 Latin → U+0441 Cyrillic small letter es
    'x': '\u0445',  # U+0078 Latin → U+0445 Cyrillic small letter ha
    'y': '\u0443',  # U+0079 Latin → U+0443 Cyrillic small letter u
    # ... 12 additional pairs (19 total)
```

```
}

def apply_homoglyphs(text: str) -> str:
    """Apply all HOMOGLYPH_MAP substitutions to a text string."""
    if text is None: raise TypeError('text must not be None')
    return ''.join(HOMOGLYPH_MAP.get(ch, ch) for ch in text)

def normalise(text: str) -> str:
    """Reverse: collapse Cyrillic homoglyphs back to ASCII equivalents."""
    reverse_map = {v: k for k, v in HOMOGLYPH_MAP.items()}
    return ''.join(reverse_map.get(ch, ch) for ch in text)
```

5.7.3 URL-Safe Attack Generator (adversarial/attack_generator.py)

[Figure 5.4: apply_homoglyphs_to_url() — the implementation of the URL-safe attack scope constraint described in Section 4.10.1.]

```
# adversarial/attack_generator.py
from urllib.parse import urlparse, urlunparse
from .homoglyph_map import apply_homoglyphs

def apply_homoglyphs_to_url(url: str) -> str:
    """
    Apply homoglyph substitution ONLY to netloc and path components.
    Scheme (http/https) is NEVER modified — Cyrillic in scheme
    produces syntactically invalid URLs rejected by all URL parsers,
    making the attack trivially detectable and scientifically invalid.
    """
    if url is None:
        raise TypeError('URL must be a non-None string')
    if not isinstance(url, str):
        raise TypeError(f'Expected str, got {type(url).__name__}')
    if not url:
        return url
    parsed = urlparse(url)
    return urlunparse((
        parsed.scheme,              # NEVER attacked — scheme preserved
        apply_homoglyphs(parsed.netloc),   # domain attacked
        apply_homoglyphs(parsed.path),     # path attacked
```

```
        parsed.params,              # NEVER attacked
        parsed.query,               # NEVER attacked
        parsed.fragment             # NEVER attacked
    ))
```

5.7.4 GiniExplainer — Unified XAI Interface (explainer.py)

[Figure 5.5: GiniExplainer class implementing the unified explain() interface across Gini, SHAP, and LIME. All methods return real feature names sourced from FeatureExtractor.get_feature_names().]

```
# explainer.py
import shap, time
from lime.lime_tabular import LimeTabularExplainer

class GiniExplainer:
    def __init__(self, model, feature_names: list):
        self._model        = model
        self._feature_names = feature_names

    def explain_gini(self, n_top: int = 5) -> dict:
        """Gini attribution: zero inference-time cost. Global importance."""
        importances = self._model.feature_importances_
        ranked = sorted(zip(self._feature_names, importances),
                        key=lambda x: x[1], reverse=True)
        return dict(ranked[:n_top])

    def explain_shap(self, feature_vector) -> dict:
        """SHAP TreeExplainer: ~0.23ms. Local (sample-specific) importance."""
        explainer   = shap.TreeExplainer(self._model)
        shap_values = explainer.shap_values(feature_vector.reshape(1,-1))
        return dict(zip(self._feature_names, shap_values[1][0]))

    def explain_lime(self, feature_vector, training_data) -> dict:
        """LIME TabularExplainer: ~130ms. High-quality local; batch/offline use."""
        lime_exp = LimeTabularExplainer(
            training_data, feature_names=self._feature_names, class_names=['legit','phish']
        )
```

```
    explanation = lime_exp.explain_instance(
        feature_vector, self._model.predict_proba, num_features=5, num_samples=500
    )
    return dict(explanation.as_list())
```

5.7.5 SHA-256 Model Integrity Verification (experiments/baseline_eval.py)

[Figure 5.1 (continued): Model integrity verification function with explicit scope limitation documentation.]

```
# experiments/baseline_eval.py
import hashlib, pathlib

def verify_model_integrity(model_path: str) -> bool:
    """
    Verify SHA-256 checksum of model file before loading.
    SCOPE: Detects file corruption and accidental modification.
    LIMITATION: Does NOT protect against an adversary with simultaneous
    write access to both model_path and model_checksum.sha256.
    This limitation is intentionally documented per EU AI Act transparency.
    """
    checksum_path = pathlib.Path(model_path).with_suffix('.sha256')
    stored_hash   = checksum_path.read_text().strip()
    with open(model_path, 'rb') as f:
        computed = hashlib.sha256(f.read()).hexdigest()
    if computed != stored_hash:
        raise RuntimeError(f'Model integrity check FAILED: {model_path}')
    return True
```

5.8 Summary

This chapter has provided complete implementation detail for SentinEL's five core components. Classifier selection and hyperparameter configuration (Section 5.1) produced a Random Forest model achieving 97.2% baseline accuracy with a 5-fold cross-validation variance of ±0.4%. Baseline performance analysis (Section 5.2) contextualised the accuracy figure through an operational error analysis distinguishing false positive and false negative costs. Adversarial

attack implementation (Section 5.3) established the 15.8pp accuracy degradation under 19-pair homoglyph substitution attack, with a feature sensitivity diagnostic identifying lexical features as the primary adversarial attack surface. Explainability benchmarking (Section 5.4) empirically established that Gini (< 0.001ms) and SHAP (~0.23ms) meet the 2ms SOC real-time threshold while LIME (~130ms) is limited to batch/offline use. The operational pipeline (Section 5.5) achieves 5× throughput improvement via ThreadPoolExecutor concurrency. The test suite (Section 5.6) achieves 10/10 passing tests in 1.82s with GitHub Actions CI. Five key code implementations (Section 5.7) demonstrate the implementation logic for the most architecturally significant design decisions.

Chapter 6: Conclusion and Future Work

6.1 Conclusion

SentinEL has achieved all stated research objectives and contributes three empirically-grounded, independently verifiable findings to the phishing detection literature.

The first and primary finding is that high-performing ML phishing detectors are materially vulnerable to Unicode-based evasion. A Random Forest classifier achieving 97.2% baseline accuracy — performance comparable to published state-of-the-art for URL-based phishing detection — degrades to 81.4% accuracy under a systematic 19-pair homoglyph substitution attack. The 15.8 percentage point degradation represents approximately 348 additional misclassifications on the 2,200-URL test set, the overwhelming majority of which are false negatives — missed phishing URLs. This finding has an immediate practical implication: any URL classifier trained without adversarial data augmentation, Unicode normalization preprocessing, or adversarial robustness evaluation is potentially operating with a significant hidden vulnerability that standard clean-input benchmarking will never reveal. The fact that no published phishing detection paper reports adversarial robustness figures underscores that this vulnerability is not merely possible but is a systematic gap across the entire research and deployment landscape.

The second finding concerns the structural nature of adversarial sensitivity in the 17-feature vector. The feature sensitivity diagnostic unambiguously identifies string-level lexical features — Shannon entropy and keyword frequency — as the primary adversarial attack surface. Metadata features — domain age, SSL validity, IP-as-host, HTTPS protocol, URL shortener detection — show zero sensitivity because they are independent of URL string character composition. This distinction provides a concrete, empirically validated design principle for adversarially robust phishing classifiers: weight metadata features more heavily relative to lexical features, because metadata constraints represent economic and temporal attacker barriers that cannot be overcome through string manipulation alone. An attacker can substitute 'a' for its Cyrillic equivalent trivially; they cannot fake a domain registered three years ago.

The third finding establishes the real-time explainability landscape for SOC deployment. Gini Impurity attribution (< 0.001ms) and SHAP TreeExplainer (~0.23ms) both meet the 2ms operational threshold; LIME TabularExplainer (~130ms) does not. The operationally significant distinction is not merely latency: Gini provides global feature importance (the same ranking for every prediction) while SHAP provides local feature importance (different attribution per URL). For SOC analysts needing to understand why a specific URL was flagged as phishing, SHAP's local attribution is the more informative and therefore the more deployable choice — despite being approximately 230× slower than Gini, it remains well within the real-time threshold. This finding provides directly actionable guidance for XAI tool selection in security operations, addressing a gap in the published literature identified in Section 1.5.6.

6.2 Theoretical and Practical Implications

Theoretically, SentinEL's adversarial evaluation contributes to the growing body of work on adversarial ML in cybersecurity applications (Apruzzese et al., 2022; Liang et al., 2016). The 15.8pp degradation under homoglyph substitution establishes the first published quantitative baseline for this specific attack class against URL-based Random Forest classifiers — a benchmark against which future adversarially-robust phishing detectors can be measured and compared.

The feature sensitivity principle — that lexical string features are systematically more adversarially exploitable than metadata features — is applicable beyond URL classification. Any ML system using text string features in a security context faces analogous Unicode-based evasion risks. The design principle applies to email header analysis, domain generation algorithm (DGA) detection, and social engineering pattern recognition — all domains where adversarial actors can control the string content of inputs.

Practically, the mitigation for the identified vulnerability is well-defined and implementable: Unicode NFKD (Normalisation Form Compatibility Decomposition) preprocessing applied to URL strings before feature extraction collapses all 19 Cyrillic homoglyph pairs to their canonical ASCII equivalents. The normalise() function implemented in adversarial/homoglyph_map.py is the exact inverse of the attack function — it constitutes the defensive preprocessing step. SentinEL's adversarial evaluation scripts are directly repurposable as a regression test suite: after

implementing NFKD preprocessing in feature_extractor.py, running adversarial_eval.py should produce attacked accuracy approaching or equalling baseline — providing continuous quantitative verification that the defence remains effective as the classifier is updated.

6.2.1 Positioning Against Published Baselines

Table 6.1 positions SentinEL's results against the closest published baselines in the feature-engineered ML phishing detection literature.

Reference	Method	Dataset	Accuracy	Adversarial Eval	Features
Jain & Gupta (2019)	Random Forest	UCI Phishing Dataset	97.7%	None	30 (URL + HTML)
Mohammad et al. (2015)	Random Forest	UCI Phishing Dataset	97.1%	None	30 (URL + HTML)
Marchal et al. (2014)	PhishStorm Classifier	PhishStorm Dataset	94.0%	None	36 (URL + DNS)
SentinEL (this work)	Random Forest	PhishTank + legitimate corpus	97.2% clean / 81.4% attacked	YES — 15.8pp degradation	17 (URL-only)

Table 6.1: SentinEL compared to published URL-based phishing detection baselines. Two observations: First, SentinEL's 97.2% baseline accuracy is competitive with the published field despite using 43% fewer features than the closest baselines (17 vs. 30 features), suggesting that the selected 17 features capture the core discriminative signal effectively. Second, and more significantly, SentinEL is the only system in this comparison that evaluates and reports adversarial robustness — the 81.4% attacked accuracy figure has no comparable published baseline because the benchmark simply does not exist in the prior literature. SentinEL's adversarial evaluation is therefore not only a contribution to knowledge but the establishment of a new evaluation standard that future work must adopt to be considered complete.

For the explainability latency comparison: Ribeiro et al. (2016) report LIME latency in seconds in their original paper on complex tabular datasets. SentinEL's 130ms measurement on a 17-feature input represents an optimistic best case; on larger feature vectors or with N > 500 neighbourhood sampling, LIME latency would be significantly higher. This contextualises the finding: LIME is challenging even for small feature vectors and would definitively fail the 2ms threshold for any realistic production phishing detection system with richer features.

6.3 Limitations and Challenges

Limitation	Technical Challenge	Proposed Mitigation
Static Training Dataset	11,000-URL dataset is a fixed temporal snapshot. Phishing tactics evolve; the trained model will degrade as attackers adapt to the known feature set.	Continuous training pipeline: deploy the FEEDBACK table as the data pipeline for periodic retraining triggered by analyst-verified misclassifications.
Single Attack Class	Only Latin-to-Cyrillic homoglyph substitution evaluated. IDN homographs from Arabic, Hebrew, Greek, and Armenian scripts not covered; structural attacks (subdomain padding, query-string obfuscation) not evaluated.	Expand HOMOGLYPH_MAP to multi-script coverage; add structural attack generators; measure degradation per attack class in a comprehensive adversarial taxonomy.
No Normalization Baseline	The NFKD normalization defense is identified and implemented as	Implement NFKD preprocessing in feature_extractor.py and re-run adversarial_eval.py — expected to restore near-97.2% attacked accuracy.

	normalise() but not evaluated — the report does not measure how much accuracy recovers under the defense.	
Demo Dataset Scope	The 14-row demo_sample.csv shipped in the repository cannot reproduce the 15.8pp degradation result. A researcher cloning the repo without the full dataset will see different results.	Add download script to README.md for the full 11,000-URL research corpus; document this requirement explicitly in the reproduction instructions.
SHA-256 Scope Limitation	The model integrity check does not protect against simultaneous modification of both the model file and checksum file by an adversary with repository write access.	Production deployment should use a Hardware Security Module (HSM) or remote attestation service to anchor the trusted checksum outside the model's filesystem.
Audit Database Persistence	ShardManager operates in-memory only — process restart loses in-memory state.	Persist shard assignments to the QUERY table in the audit ledger; implement session recovery on process startup

6.4 Future Work

6.4.1 Adversarial Robustness Extension

12. Unicode Normalization Defence Evaluation: implement NFKD preprocessing in the feature extraction pipeline and measure the recovery in attacked accuracy. This is the

most immediate next experiment — it directly quantifies the effectiveness of the identified mitigation and closes the loop between vulnerability discovery (Chapter 5) and defensive implementation.

13. Multi-Script Homoglyph Attack Taxonomy: extend HOMOGLYPH_MAP to cover Arabic, Hebrew, Greek, Armenian, and Georgian confusables (as catalogued in the Unicode Consortium's Confusables database). Evaluate degradation per script family to identify which scripts pose the greatest practical evasion risk against the Random Forest feature vector.
14. Adversarial Training: augment the training dataset with homoglyph-attacked examples and measure whether adversarial training improves robustness without degrading clean-input accuracy — the standard adversarial training evaluation methodology.

6.4.2 Explainability and Transparency Enhancement

15. Case-Based Reasoning Integration: implement a CBR-based explanation system that retrieves the most similar previously-classified URLs as supporting justification for new predictions. This extends SHAP's numerical attribution with human-interpretable precedent examples — directly aligned with the CBR explainability research conducted at UCD ML-Labs and providing a concrete connection between this undergraduate research and the proposed MSc Computer Science research agenda at UCD.
16. SHAP Interaction Values: extend the benchmark to measure pairwise feature interactions via shap_interaction_values(), identifying which feature combinations are jointly predictive and whether adversarial attacks disrupt learned interaction patterns.
17. EU AI Act Article 13 Compliance Audit: generate a formal transparency report for the SentinEL classifier documenting: model capabilities and limitations, available explanation mechanisms and their latency characteristics, adversarial robustness measurements, and recommended human oversight protocols — directly implementing the Article 13 technical documentation requirement.

6.4.3 Operational and MLOps Enhancement

18. Live URL Feature Extraction Service: extend the Autonomous Feature Extraction layer to perform HTTP/WHOIS/DNS lookups at classification time, enabling metadata features (domain age, SSL validity) to be extracted from live URLs rather than pre-computed values in the dataset. This would eliminate the current requirement for pre-computed metadata while increasing the classifier's coverage to any URL submitted in real-time.
19. Continuous Training Pipeline: implement the full FEEDBACK → retraining loop: analyst-verified misclassifications accumulate in the FEEDBACK table, a scheduled retraining job triggers when the feedback count exceeds a threshold, the new model is evaluated against the adversarial test suite before promotion to production, and SHA-256 checksum and MODEL_VERSION table are updated atomically.
20. SIEM Integration Connectors: develop connectors for Splunk (SPL custom command) and IBM QRadar (Custom Properties + AQL query) — the two dominant SIEM platforms in the Irish financial services sector — enabling SentinEL to operate as a threat intelligence enrichment plugin with SHAP attribution surfaced in analyst dashboards alongside standard IOC indicators.

6.5 Summary

SentinEL presents three independently verifiable contributions to the machine learning-based phishing detection field. The primary contribution — quantifying 15.8pp accuracy degradation under systematic Unicode homoglyph attack — demonstrates that adversarial robustness evaluation is not optional for security-critical ML systems. The secondary contribution — empirically establishing real-time XAI latency thresholds for Gini (< 0.001ms), SHAP (~0.23ms), and LIME (~130ms) — provides directly actionable SOC deployment guidance absent from the existing literature. The tertiary contribution — the modular, fully-tested, SHA-256 verified, CI-enabled codebase — provides a reproducible research artifact suitable as a foundation for graduate-level adversarial ML research.

The project was developed collaboratively by Mourya Reddy Udumula (dataset curation, ML architecture, adversarial evaluation, explainability benchmarking) and Jeet Upadhyaya Anand (operational pipeline, concurrent IOC ingestion, forensic audit logging). Findings were presented at the Indrashil University Annual Research Symposium 2025.

The single most important methodological contribution is not a specific result but a principle: evaluating a security ML classifier on clean inputs alone is insufficient. The adversarial evaluation pipeline developed in SentinEL — reusable, modular, and parameterised — constitutes a general-purpose tool for adversarial robustness measurement that the broader URL-based phishing detection community can adopt to close the systematic evaluation gap identified in the literature review.

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